

Quiz #4

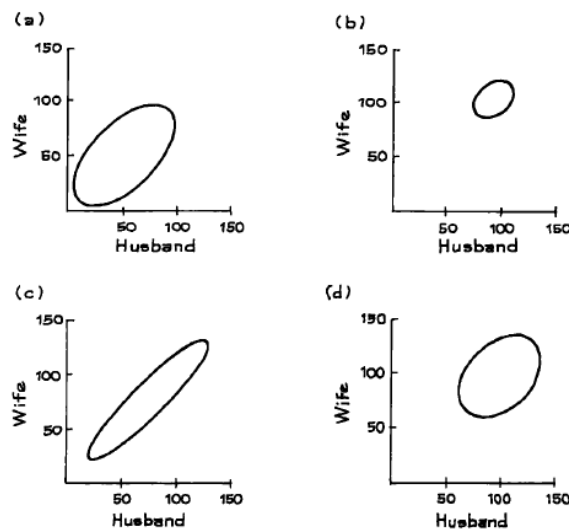
1. A laboratory repeatedly measures a reference mass whose exact value is known to be 50.00 g. Five measurements are

50.08, 50.07, 50.09, 50.08, 50.08 g.

- (a) Are these measurements precise? Briefly explain.
 - (b) Do the measurements appear to be accurate? What type of measurement error is suggested by these readings?
 - (c) Would taking many more measurements and averaging them eliminate this problem? Briefly explain.
2. A study of the IQs of husbands and wives obtained the following results:

$$\bar{x} = 100 \quad (\text{husbands}), \quad \bar{y} = 100 \quad (\text{wives}),$$

$$s_x = 15, \quad s_y = 15, \quad r = 0.6.$$



Which scatter diagram matches these data? Briefly explain why you reject the others.

3. Suppose `gap_2007` contains the Gapminder observations from 2007. Consider the following code:

```
result = gap_2007.loc[
    (gap_2007["continent"] == "Americas")
    & (gap_2007["pop"] > 10_000_000),
    ["country", "lifeExp", "gdpPercap"],
].sort_values("lifeExp", ascending=False)
```

- (a) In words, which rows are selected?
- (b) Which columns appear in `result`, and how are the rows ordered?
- (c) Why is `&` used instead of `and` when combining the two conditions?